

Designing an Autonomous Integrated Downstream Sequence From a Batch Separation Process — An Industrial Case Study

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This work is a proof of concept of how a sequence of industrial batch separation steps together are used to form an integrated autonomous downstream process. The sequence in this case study consisted of an anion chromatography step, virus inactivation and finally a hydrophobic chromatography step. Moving from batch to integrated separation minimizes hold-up times, storage tanks, and required equipment. The conversion from batch to integrated mode is achieved by extracting operating points and separation data from batch chromatograms. The integrated separation process is realized on an ÄKTA Pure controlled by an open research software called Orbit, making it possible to operate complex process configurations including multiple steps. The results from this case study is the principle and method of the steps taken to automation, achieving a more continuous and efficient downstream process.

1. Introduction

Increasing demands on biopharmaceutical drugs requires a progressive approach of bioprocesses to become more productive and efficient.^[1] Continuous upstream processes require an integrated approach for handling a constant feed. There are many advantages with an integrated sequence, among them are the reduced risk due to less manual handling, reduced cycle time, and less equipment.^[2,3] There are also substantial economic advantages in using an integrated continuous biomanufacturing platform.^[4] Combining this with the flexibility of managing different products in the same facility, could make the implementation very profitable^[3] as well as environmentally friendly.^[5,7] The U.S. Food and Drug Administration is supporting the implementation of continuous processing.^[6]

There are studies showing the productivity and flexibility gain as well as opening up possibilities for implementing control strategies in continuous and integrated downstream processes. The footprint is generally reduced with a continuous approach since the

equipment and investment is substantially decreased.^[7] Sellberg et al., for example, showed that antibody dimerization in downstream processes were less pronounced in integrated production compared to batch wise production.^[8] In other industries, such as the U.S. steel industry, there were many large companies that did not adapt to continuous production and they were eventually outcompeted.^[9] This is a reason for biotech companies to start implementing continuous processes.

The main focus of previous studies has been on Periodic Counter Current Chromatography (PCCC), Multicolumn Countercurrent Solvent Gradient Purification process (MCSGP) and Simulated Moving Bed (SMB).^[7] The PCCC technique is an

effective loading step to capture the maximum amount of product from the harvest.^[10] In SMB, the size differentiation or other characteristics of two similar components, is what this technique utilizes for separation, in a single unit operation.^[11] MCSGP is also a single unit operation that effectively separates product from impurities and recycling the part where the product is not sufficiently pure.^[12] There are also simpler and more straightforward methods for connecting the upstream to the downstream. One such has been performed with a perfusion cultivation. With two alternating capture columns working in 24-hour cycles the downstream becomes semi-continuous with the alternating release of the product to the next step.^[13]

The advantage of the integrated column sequence is that it can use a variety of different purification steps on a single system with reduced manual steps and holdup times. Integrated column sequences also offer a way to take advantage of a continuous capture step, and this ultimately creates a truly end-to-end production and purification process including a number of unit operations. It is also possible to implement the integration of cell lysis and protein re-folding in a continuous production.^[7,14]

A white paper on continuous bioprocessing emphasizes that a higher degree of automation is associated with cost savings, less manual labor and operational consistency due to less subjective decision making. The paper also comments on the different levels of maturity in continuous upstream and downstream processes, where the downstream technologies is lagging and would benefit from further research and experience.^[15]

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Optimized time lines and regulations challenge the manufacturing process which is costly to change after the preclinical and clinical trials, with a comparability study often being required. A new application might be required if the process is moved out of the design space.^[16,17] This would considerably increase costs. This is why the greatest benefit of implementing integrated and continuous processes lies already in the early stages of testing.

A couple of studies have implemented a scripting approach testing novel control strategies.^[18,19] This work extends the technique on an integrated column sequence (ICS) with an industrial case study that reduces several manual steps by making them automated. An existing downstream process contained three clarifying filter steps followed by a chromatography step, a virus inactivation step, an additional chromatography step and finally a few filtration steps. The aim of this case study is to reproduce a part of this process, the two chromatography steps and the virus inactivation, on a single unit in an ICS to show the proof of concept.

2. Experimental Section

An ÄKTA Pure 150 from GE Healthcare (Buckinghamshire, United Kingdom) was used as a building platform. On that platform, in combination with UNICORN 7, an OPC server was used to communicate with the hardware. This was done via the general purpose, object oriented scripting language Python (<http://python.org>).

Note that *steps* in this paper is used to describe different unit operations, such as a column step or VI step, while *phases* is used within the unit operations, such as wash, load, elute phases et cetera.

2.1. Materials

The basic setup of the ÄKTA Pure was a column valve and a loop valve. The columns HiTrapTM DEAE FF 5 ml (AIEX) and Phenyl FF High Sub, two coupled 5 ml, were used. A superloop of 150 ml was used for the virus inactivation (VI) step, connected to the column valve in the same fashion as the columns. There was also a connected sample pump, S9, used for the inline dilution. PEEK tubing with an inner diameter of 0.75 mm connected the system. The batch experiments used the Fraction collector F9-C.

The drug substance was a recombinant human protein expressed in Chinese hamster ovary cells, CHO, in the facilities of Swedish Orphan Biovitrum AB. The cell harvest was clarified by depth filtration followed by 0.45/0.2 µm filtration. Before being applied to the downstream process, the solution was diluted with a dilution buffer, 1:1.2.

In total there were six different buffers used in the AIEX step and six different in the HIC step, including the dilution buffers. The pH range was between 4 and 14.

2.2. Experimental Setup

The two columns were connected to the loop valve and the superloop on the column valve, see **Figure 1**. Before and after these valves there were two versatile valves to select to which flow

path to use. The detectors (pH, conductivity and UV) were placed after the second versatile valve.

After the detectors, the outlet valve was placed to either direct the flow to waste or back to the first versatile valve and to the next step. Between the outlet valve and the first versatile valve, a T-cross was placed to allow dilution from the sample pump. The column valve had the ability to reverse the flow which allowed the option of emptying the superloop.

The three-way junction, which can be seen in Figure 1, allowed the sample pump to dilute the pools, of the first two steps, which is returned to the first versatile valve. The dashed line shows the path from A buffers to waste via the AIEX column. When eluting, the pool was directed in the outlet valve to the first versatile valve with dilution and sent to the superloop.

The AIEX buffers were all placed on inlet valve A and the HIC buffers on inlet valve B. The reason is that the AIEX buffers are similar to each other and HIC buffers to each other. The effect of residues in the tubing in the two inlet valves would thus be reduced. The in-line dilution buffer was connected to the sample pump valve.

The connection to the versatile valves allowed the flow path to pass them twice, simultaneously. The inlets were connected to port 1 and 3, and the outlets were connected to port 2 and 4. The versatile valves were either in the setting 1 to 2 and 3 to 4, or in the setting 1 to 4 and 2 to 3. With this configuration, the flow path was directed to either the loop valve or the column valve and when the pool was sent back from the outlet valve, it would automatically be directed to the other valve, that is, loop or column.

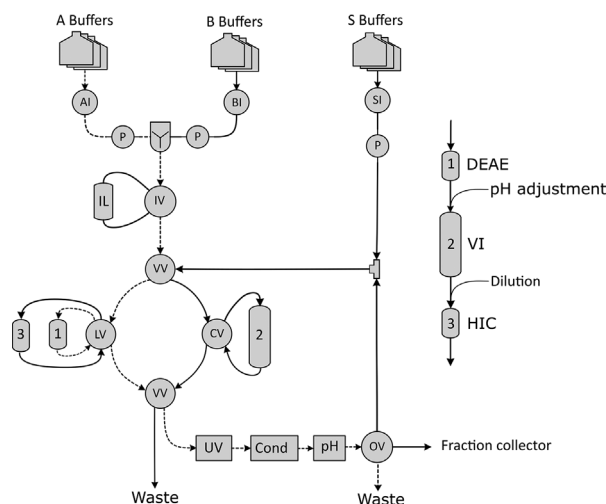


Figure 1. The experimental platform. From the top, the buffers are connected to the buffer inlets, AI and BI, pumped with two pumps, P, to the mixer and together down through the injection valve, IV, that has an injection loop, IL. The first versatile valve, VV, directs the flow to either the loop valve or to the column valve. The two columns are placed on the loop valve, LV, while the superloop is on the column valve, CV. The second versatile valve directs the flow to the detectors and then to the outlet valve, OV, where it is either sent to the waste, the fraction collector or back to the first versatile valve where it is directed to the next step. The dashed line represents the pathway of the first step, the AIEX column.

2.3. Experimental Methodology

The first experiments were conducted with separate batch steps while the later was the complete sequence. In the separate batch experiments, the operating set points provided by Swedish Orphan Biovitrum, SOBI, were extracted and used to replicate the process in our facilities. When developing the ICS, the column volumes and phase volumes (load, wash, elute, etc.) were kept constant.

During the virus inactivation step the entire tube system was cleaned with the HIC wash buffer. This was performed to allow the inactivated pool adsorb to the HIC resin, minimizing any residuals from the AIEX step to have an effect on the loading phase.

The separate batch experiments were a scale-down replication, keeping consistent linear flow rates and phase times (in column volumes). Both the experiments in separate batches and the integrated sequence were performed with a load of 27 ml filtered and diluted cell harvest.

The design of the second column was used as a base to scale down the sequence, it was chosen to be 10 ml. The first column volume was then constrained to its loading capacity and had to be calculated. The pool from the first column step was directly loaded into the virus inactivation step. The dilution after the VI step started when the inactivated pool was calculated to arrive at the T-cross, via the outlet valve. The calculation was based on the flow rate and the dead volume between the superloop and the T-cross. The dilution ended when the superloop was empty. The inline dilution factor was 1:0.8.

2.4. Experiments

First, batch experiments were performed that was a scale-down from clinical experiments. The clinical experiments are not discussed in this paper, only the batch and the integrated experiments.

All experiments performed in this study were performed with a load of 27 ml that was pre-diluted 1:1.2 with a dilution buffer. This corresponds to 20% of the reference scale experiments. The virus inactivation step was simplified. The pH-adjusted pool from the first AIEX column was stored in a superloop for a shortened time period before loaded onto the second HIC column.

In the AIEX experiment, the column is first equilibrated and then it is loaded with the filtered drug solution. The HIC step starts with an equilibration phase followed by the loading phase and three wash phases, w1, w2 and w3. After that the eluting phase and then the final pool. Finally, the column is cleaned.

A pool flow rate of 0.11 column volumes per minute from the 5 ml AIEX column and a load flow rate of 0.1 column volume per minute on the 10 ml HIC column is used. The integrated experiment is performed using a five times higher flow rate than the batch experiments, while keeping the same volumes in the phases. After the VI step, the pool is eluted and loaded onto the last column in a similar fashion to the pool from the AIEX column, except that the dilution is activated.

Since the HIC column was eluted with a low salt concentration, the system is filled with the low salt washing buffer before

emptying the superloop. This is done in a way that no residual buffers remain in the tube system and so that the pool adsorbs to the HIC resin and not flow through with the residual buffers. This is performed during the virus inactivation phase in the integrated sequence.

2.5. Controller

All experiments were performed through Orbit which was developed at the department of Chemical Engineering at Lund University. It works on top of UNICORN through OpenOPC, which is an OPC (Object Linking and Embedding for Process Control) toolkit designed for use in the programming language Python.

The controller was first connected to the machine server. When contact was established, pumps and valves could be controlled. Through the OPC connection, there was full access to writing (i.e. change valves and pumps) and reading (e.g., UV, conductivity, pH and pressures).

Orbit is built in Python and has mapped the rudimentary instructions from the OPC interface to simple instructions inside the program, such as the example *system.pump.setFlowrate(2)* which sets the flow rate to 2 ml min⁻¹. This allows the implementation of complex methods with high level input. The basic way is letting the user set the instructions to all valves and pumps at specific times while the higher level is letting the user give inputs that utilize the information that orbit has about the system, e.g. volumes and configurations, and automatically define instructions and timings based on that. The high level input could e.g. be setting a column, duration of phases and length of the pool. All these inputs are then decoded, analyzed and used to control the system.

It is possible to supervise other hardware from orbit provided they can be controlled through OPC. Andersson et al. (2017), has discussed in depth Orbit's possibilities and limitations.^[19]

3. Results and Discussion

The results of this case study is focused on the methodology of the integration, how the system was set up and the consequences of those choices. Therefore, the results are mainly qualitative and not quantitative.

3.1. Design

The 10 ml HIC column was used as a base for optimization calculations. The result from this was that the AIEX column must be equal to or less than 5 ml which is why the 5 ml column was chosen. This was calculated based on the capacity and the volume of the columns so that the HIC column would not be overloaded.

The VI step separates the flow path between the two column steps. This means that there is no direct transfer of the pool from column to column and the flow rate can be freely tuned to each column. The setup can directly be scaled up at least 100 times before the flow rate limitation of 150 ml min⁻¹ from the system unit.

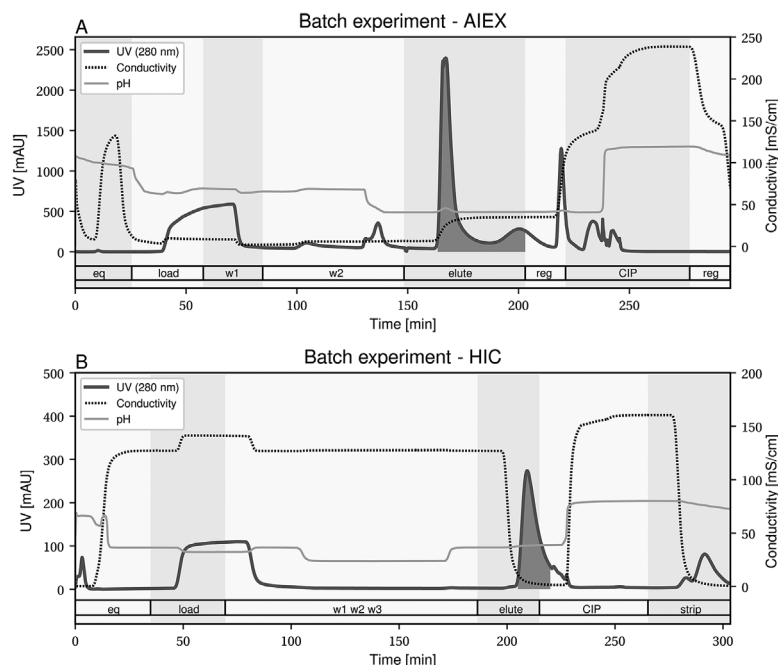


Figure 2. The batch experiment on the AIEX column is displayed in subfigure (A) and the HIC experiment in subfigure (B). These experiments were replicated from the extracted operational points from SOBI processes. The pool in both chromatograms is the shaded area under the curves. The pool from the AIEX was extracted and loaded into the HIC column. The varying gray background, as well as the text boxes below the curve, indicates the different phases.

Choosing another scaling case than the 10 ml HIC column would increase the maximum size limit of the AIEX column. While scaling up and keeping a constant flow rate in column volumes per hour, the amount of product produced per day is directly affected. For small scale manufacturing facilities, the available system hardware, for example, pumps, sensors, and tubing, will eventually be the limiting factor since the size of the columns easily can be scaled up. However, with an efficient use of this equipment, the required amounts for preclinical and early clinical trials can more readily be met.

3.2. Separate Batch Experiments and the Integrated Column Sequence

The separate batch experiments are compared in **Figure 2** and the results of the integrated sequence are shown in **Figure 3**. The boxes below the plots represents the time when the different columns and the superloop are active, enhanced by the alternating gray background. The shaded areas under the curve are the pools. In the ICS case, since the elute phase of one step occurs simultaneously as a load phase in a following step, two phases are reduced to one. This occurs in two instances in this case study, before and after the VI step.

An effect of the superloop is that the pool coming from that step is the reversed peak as that from the first column. The integrated sequence reduces the holdup time of the protein since the regeneration phases of the first column step are held up and performed later. The total holdup is reduced to the time between the loading phase and the third eluting phase, also combining pool phases with load phases in the AIEX column to the VI and the VI to the HIC column. An alternative would be to clean the AIEX column during the virus inactivation to decrease the total time, if there is sufficient time in between.

3.3. Productivity

With the extinction coefficient, the concentration in the pool was measured with the UV signal and calculated. The integrated sequence produced roughly 16 mg of product with a total cycle time of 13.6 hours, including a VI time of 60 min, loading the column to capacity maximum and a flow rate which is the same as the batch experiments. Since every phase is equal to the separate batch experiments, the productivity would in theory appear to be equal, which they in reality are not due to the automation.

It is difficult to quantify the amount of reduced manual work using the integrated approach since there is a lot of factors varying across companies and individuals. Although, it saves manual labor as well as time in at least seven steps. First, there is the collecting of the pool from the first column step, then there is the transfer to the virus inactivation step, performing the pH adjustment and then the dilution. Finally, there is the loading to the second column step. Loading the first column and collecting from the second column were performed both in the ICS experiment and the separate experiments. In this industrial case study, 7 steps became reduced to 2. Loading once and collecting once.

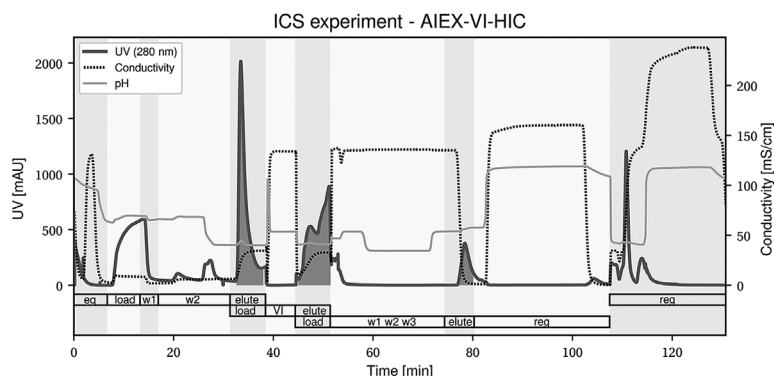


Figure 3. The complete integrated column sequence. The duration of the different phases is represented by the boxes under the graph and the level indicates the step; AIEX, VI, or HIC. They are also enhanced by the alternating gray background. UV, conductivity and pH is shown continuously over all three steps. The regeneration phase of the first column is placed last.

Considering that the all steps are automated, the experiments could be performed overnight. The controller also makes it possible to run the system automatically, allowing as many experiments as needed to be run in sequence, without any manual interference.

3.4. Conclusion

The main contribution of this work is the methodology for applying an industrial downstream process to a single unit integrated column sequence. For small scale production, process development or research, this is an interesting option due to the efficient use of equipment and setup. With the chosen column sizes 5 and 10 ml, the process can be scaled up at least 100 times on the system used in this case study.

The method performed in this work showed that it was possible to take an industrial case that was used in clinical trials and integrate into a single unit, producing 16 mg of product every 13.6 h. For small scale manufacturers, the available equipment and system hardware will eventually be a limiting factor which is the reason that finding the most effective method for producing therapeutics will become so important. The efficiency was shown by how it is automated and avoids several manual steps, reducing 7 manual steps to 2, thus saving manual labor due. The automation allows overnight experiments as well as the possibility to run multiple experiments in sequence.

Abbreviations

ALEX, anion exchange; CIP, cleaning in place; HIC, hydrophobic interaction chromatography; ICS, integrated column sequence; MCSGP, multicolumn countercurrent solvent gradient purification; PAT, process analytical technology; PCCC, periodic counter-current chromatography; PEEK, polyether ether ketone; SMB, simulated moving bed; VI, virus inactivation.

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Conflict of Interest

The authors declare no commercial or financial conflict of interest.

Keywords

chromatography, continuous bioprocessing, industrial case study, integrated column sequences, multiple column control

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